CS528 Homework 5

Cloud SQL Request Logging and Operation Timing

Yide Fang | Boston University | October 23, 2026

Code: https://github.com/ThomasFYD/CS528-HWs
Directory: **CS528-HW5**

1. Design and deployed resources

I extended the HW4 file service to persist request metadata, unsuccessful responses and operation timings in Cloud SQL. Two instructor clients run concurrently on my Mac and send requests to a Web VM. Allowed requests read original files from the HW2 Cloud Storage bucket; forbidden-country requests publish to Pub/Sub for a separate Reporter VM.

Component	Actual configuration
Project	dazzling-seat-508219-j3
Region / zone	us-central1 / us-central1-a
Web VM	e2-small; Debian 12; up to 8 request workers
Reporter VM	e2-micro; Debian 12; unchanged HW4 reporter
Cloud SQL	MySQL 8.4 Enterprise; db-g1-small; 10 GB SSD
Database / instance	hw5 / cs528-hw5-mysql
Storage / HTTP	cs528-hw2-yf-2026; data/0.html to data/11999.html; port 8080
Guard	Authenticated generation-2 function; hourly at minute 5 UTC

Application behavior and HW4 continuity

GET and POST retain the HW4 file-serving behavior. Missing objects return 404, unsupported methods return 501, and the nine assignment countries return 400. The country check uses the simulated X-country header. No application object cache is used. Each handled request commits its metadata and any error row before the response is sent; a logging failure returns 503 and is recorded as an operational error.

The second application, reporter.py, is byte-for-byte unchanged from HW4. It pulls forbidden-country events, prints them and acknowledges them. The verified SHA-256 is:

`095e77c433957dd45185d4c0fce3efa3a948866308839bdabb9c22dbb93c87b0`

The completed formal experiment contains **100,000 requests**: 95,653 HTTP 200 and 4,347 expected forbidden-country HTTP 400 responses. Both clients exited normally. Cloud SQL was retained and stopped after the experiment; the temporary VMs and web address were removed.

2. Configuration and execution

Run these commands from the homework directory on the Mac used for the experiment. Use an authorized Google Cloud identity and the updated macOS instructor client. config.py hardcodes dazzling-seat-508219-j3; cloud commands explicitly specify that project rather than reading the operator's default project.

```
gcloud auth login fangyd@bu.edu --project=dazzling-seat-508219-j3
gcloud components install beta --quiet
mkdir -p client
cp ~/Downloads/http-client-mac client/http-client-mac
chmod u+x client/http-client-mac
bash verify_local.sh
bash setup.sh --plan
bash setup.sh
```

Local verification creates a virtual environment, checks syntax, runs offline tests and prints the resource plan. Setup creates or starts the owned SQL instance and provisions the HW5 deployment. It preserves existing data. Startup completion must be checked before requests are sent; the live Web and Reporter startup logs were verified in this deployment.

Principal	Runtime access
Web service account	Cloud SQL Client; project log writer; this DB password secret; object-prefix read; forbidden-topic publish
Reporter service account	Subscriber access to the forbidden-country subscription
Guard service account	Custom role with cloudsql.instances.get and cloudsql.instances.update
Scheduler service account	Authenticated invocation of the guard HTTP endpoint

VM initialization and database access

All internal VM setup runs in startup.sh through --metadata-from-file. The startup script installs dependencies once, uses a source-hash marker, starts systemd services on later boots, and invokes setup_schema.py to create only missing tables. It does not use SSH to configure VMs. A mismatched deployed source is handled by the explicit infra.py refresh-vms operation.

The Cloud SQL Python Connector provides encrypted connections using the attached VM identity. The database password is generated into Secret Manager. No public authorized-network range is added. The dedicated MySQL account is a homework database owner account, not a fully restricted production DML account. The local tools use gcloud login without creating or revoking ADC credentials.

Run sequence

```
.venv/bin/python tools/curl_demo.py
caffeinate -i .venv/bin/python tools/run_experiment.py --requests 100
caffeinate -i .venv/bin/python tools/run_experiment.py --requests 50000
```

The curl demonstration runs before the load tests so complete tables remain small. Each load command starts two clients. The runner pauses the SQL guard, starts SQL if needed, checks the schema, launches the clients, exports records and restores the guard. In a forced termination that bypasses cleanup, use python3 infra.py resume-guard.

3. Database schema

The updated client probe confirmed X-country, X-client-IP, X-gender, X-age, X-income and X-time. The main table has 15 columns; grouped rows below list columns that share a type. Missing descriptive fields remain NULL. When X-client-IP is absent, client_ip falls back to the actual peer address.

requests: one row per handled request

Column(s)	Type	Purpose / constraint
request_id	VARCHAR(36)	UUID primary key
request_time	DATETIME(6)	Non-null; unique UTC receipt time
run_id	VARCHAR(64)	Indexed experiment identifier
method	VARCHAR(32)	Non-null HTTP method
requested_file	VARCHAR(1024)	Non-null object path
country	VARCHAR(128)	Original X-country label
client_ip / peer_ip	VARCHAR(45), each	Simulated client IP / actual socket peer
gender / age / income	VARCHAR(32), each	Original client labels; no inferred numeric bins
is_banned	BOOLEAN	Non-null country-policy result
client_time	VARCHAR(64)	Original X-time value
time_of_day	VARCHAR(8)	Extracted HH:MM:SS
http_status	INTEGER	Non-null HTTP status

request_time is generated in UTC under a thread lock, with a one-microsecond tie adjustment and a database-seeded restart value. This uniqueness design assumes one server process. The repeating simulated X-time has no supplied time zone and is not used as a primary key.

failed_requests: exactly the three required columns

Column	Type	Constraint
request_time	DATETIME(6)	Primary key; foreign key to requests.request_time
requested_file	VARCHAR(1024)	Non-null
error_code	INTEGER	Non-null non-200 HTTP status

The request row and its error row commit atomically in one SQL transaction. Successful requests have no failed_requests row. Parameterized SQLAlchemy statements are used.

request_timings: operation measurements

Column(s)	Type	Purpose
request_id	VARCHAR(36)	Primary key; foreign key to requests.request_id
extract_ns, storage_ns, response_ns, database_ns, total_ns	BIGINT, each	Non-null integer nanosecond measurements
storage_attempted, response_sent	BOOLEAN, each	Non-null flags for read attempts and completed writes

Timing rows are inserted after response handling. The timing-row insert itself is excluded from database_ns and total_ns. Schema setup creates missing tables and checks existing column layouts; it does not drop tables or erase previous experiments.

4. Concurrent experiment and validation

Formal run: **load-20261021T075735Z-5b3f60c2**. Both supplied-client processes were launched before either was awaited. Their complete saved command arrays are identical. The historical invocation below is included as evidence; its released address must not be reused for a new deployment.

```
client/http-client-mac -d 35.193.157.51 -p 8080 -b none \
  -w __runs/load-20261021T075735Z-5b3f60c2/data \
  -n 50000 -i 11999 -r 528 -t 30
```

Measurement	Observed result
Clients / requests	2 simultaneous clients; 50,000 requests each; seed 528
Load start / finish, UTC	2026-10-21 07:57:43.556263 / 10:58:49.526978
Load elapsed / throughput	10,866.021 s (3 h 1 min 6.021 s); 9.203 requests/s aggregate
Client exit codes	0 and 0
Requests / unique request IDs	100,000 / 100,000
Unique receipt times / timing rows	100,000 / 100,000
Matching non-200 error rows	4,347; exact match on receipt time, path and HTTP code
Missing required metadata / send failures	0 / 0

The raw snapshot was independently recomputed and reconciled with the saved summary, including all six statistics and timing distributions. Run-specific routing separates these records from curl, smoke-test and unrelated public traffic. Both clients used the same seed argument, but the saved run has 100,000 distinct complete metadata tuples. I retain that observation without claiming identical generated sequences or assigning a cause.

Raw snapshot SHA-256 (10,514,300 compressed bytes):

`acfcd6cfb1d3fed0d05483197f138d6cefa74ca34f05bd35c4e6e5aa897877bc`

Real curl demonstration before the load run

Case	HTTP	Request rows	Error rows
Existing file, United States	200	1 -> 2	1 -> 1
Missing file, United States	404	2 -> 3	1 -> 2
Existing file, Iran	400	3 -> 4	2 -> 3

The initial row is an unrelated public GET that returned 404; it remains visible in every complete-table snapshot. Each curl call adds one request and only failures add an error row. Appendix A contains all nine screenshots: local replay of saved real October 21 evidence, captured by the user on October 22. These are not contemporaneous live terminal screenshots.

Before the formal run, the two-client smoke test completed 200 requests: 187 HTTP 200 and 13 expected HTTP 400, with complete matching database records.

5. Operation timing

Each required operation is implemented in a separate function. time.perf_counter_ns() measures elapsed intervals at high resolution, and SQL stores integer nanoseconds. Extraction includes header parsing, request-body validation and metadata extraction; the other stages measure GCS reads, response sending and the request/error transaction.

Stage	Function / measurement boundary
Extraction	parse_request() + extract_headers(); includes body validation
GCS access	TimedBackend.read(); includes an attempted read returning 404
Send response	FileHandler.send_result(); headers, body write and flush
Database insert	Database.save_request(); atomic metadata/error transaction

Observed durations in milliseconds

Operation	N	Mean	p50	p95	Maximum
Extraction	100,000	0.335557	0.323775	0.419734	3.079330
GCS read, all requests	100,000	48.967563	37.029859	103.702136	4821.151238
Response write / flush	100,000	39.487487	66.870686	72.105763	2134.559141
SQL request/error insert	100,000	7.335611	7.135867	9.073666	247.941632
Total measured work	100,000	102.369571	106.444684	174.890668	4828.324650
GCS, attempted reads only	95,653	51.192920	37.859784	104.974253	4821.151238

The four required means are **0.335557 ms** for extraction, **48.967563 ms** for GCS access across all requests, **39.487487 ms** for response sending, and **7.335611 ms** for SQL insertion. The attempted-read GCS mean is **51.192920 ms**.

Interpretation and denominators

The 4,347 forbidden requests skip GCS, contributing zero to the all-request storage mean. The attempted-read subset contains 95,653 operations. Percentiles use the nearest-rank definition. Maximum values preserve the observed long-tail operations; no outliers were removed.

The SQL interval covers the request/error transaction; the later timing-row insert is excluded from both database_ns and total_ns. Response timing covers server-side socket work, not round-trip latency or proof that the client received every byte. The recorded response_sent flag reports whether the server write completed.

Total server time also contains country-policy handling, Cloud Logging, Pub/Sub waits and other application work. It need not equal the sum of the four displayed stages. Experiment wall-clock time is measured independently and includes client/network pacing.

Reproducible analysis

analysis.py summarizes the run-scoped SQL export; queries.sql contains equivalent statistical queries. The saved summary includes sample counts, mean, p50, p95 and maximum for each stage. Independent recomputation from the raw gzip snapshot matched every reported timing statistic.

6. Required statistical answers

All answers use the 100,000 rows from the accepted formal run only. Successful and unsuccessful requests are included in the demographic counts. Original instructor-client labels are retained; numerical ages and incomes are not inferred from the ranges.

Question	Verified answer
1. Successful versus unsuccessful	95,653 HTTP 200 (95.653%); 4,347 non-200 (4.347%). All non-200 responses were HTTP 400.
2. Banned-country requests	4,347; every one returned HTTP 400.
3. Male versus female	Male: 49,919. Female: 50,081.
5. Most frequent age group	36-45: 12,638 requests (12.638%); no tie.
6. Most frequent income group	20k-40k: 12,784 requests (12.784%); no tie.

4. Top five countries

Rank	Country	Requests
1	Micronesia	579
2	Somalia	572
3	Philippines	566
4	Togo	566
5	South Korea	563

Philippines and Togo tie at 566 and are displayed alphabetically. Mexico follows with 562, so there is no tie at the fifth-place boundary.

Complete age and income distributions

Age group	Requests	Income group	Requests
0-16	12,437	0-10k	12,210
17-25	12,432	10k-20k	12,440
26-35	12,472	20k-40k	12,784
36-45	12,638	40k-60k	12,357
46-55	12,496	60k-100k	12,380
56-65	12,608	100k-150k	12,554
66-75	12,428	150k-250k	12,693
76+	12,489	250k+	12,582

Each distribution sums to 100,000. Neither gender, age nor income is missing in this run. The 4,347 denied requests are expected policy outcomes, not evidence of server overload.

7. Scheduled shutdown and cleanup

The authenticated hourly function checks this SQL instance and its activation policy. It returns immediately when the policy is NEVER; for a running instance it requests activationPolicy=NEVER. Transitional states use a retryable response. The experiment runner paused the schedule during active work and recorded a successful resume after the formal run.

Observed automatic stop after the formal experiment

Event on October 21, 2026	UTC time
Initial Cloud Run invocation returned 500: no available instance	11:05:00.964124
Retry submitted activationPolicy=NEVER	11:05:30.055253
Scheduler received HTTP 200 for the retry	11:05:32.670792
SQL STOP_INSTANCE completed with status OK	11:06:13.728159

The completed stop actor is the HW5 guard service account. Operation ID: 2c4be193-7882-407b-b333-130e00000032. The saved window contains 25 successful scheduler completions and one HTTP 500; no scheduled invocation appears during the formal load interval. Both the platform failure and the successful retry are retained.

Reporter subscription evidence

Metric in saved query window	Observed value
Sent-message DELTA samples, summed	4,347
Acknowledgment DELTA samples, summed	4,343
Latest sampled undelivered backlog	0 at 2026-10-22 07:58 UTC

These metrics establish subscription activity, but the four-count difference is unresolved. Individual Reporter stdout was not collected. I do not claim proof that all 4,347 individual messages were acknowledged, or assign the difference to a specific metric issue. The separate SQL export has all 100,000 requests and their matching error and timing records.

Verified final teardown

Resource	Final verified state
Cloud SQL cs528-hw5-mysql	STOPPED; activationPolicy NEVER; deletion protection enabled
Web and Reporter VMs	Both absent
Reserved static web address	Absent / released
Hourly Scheduler job	Absent

Cleanup stops SQL without deleting it. It retains the database, password secret, HW2 bucket and local evidence, while removing owned temporary resources. Source cleanup targets exact function-source object generations and retains the shared Google-managed bucket. The table lists independently checked resources; other cleanup actions are supported by successful command completion. Retained storage and secrets can still be billed.

8. Billing and evidence record

The service-grouped Cloud Billing snapshot covers **October 20-22, 2026** and was captured on October 22 at 05:41 local time. Its displayed total is **$0.00**, including **$1.61 in savings**. Appendix B preserves the full original screenshot.

Service	Usage cost	Other savings	Subtotal
Cloud SQL	$0.35	-$0.30	$0.05
Cloud Storage	$0.02	-$0.02	$0.00
Cloud Run Functions	$0.01	-$0.01	$0.00
Networking	$0.13	-$0.13	$0.00
Compute Engine	$1.09	-$1.15	-$0.05

Amounts and the dollar symbol are transcribed as displayed. The row values are rounded; use the console's $0.00 total rather than recomputing an exact gross total from visible cents. The negative Compute Engine subtotal is retained as shown.

I confirmed that I did not use gcloud to operate other projects during this date range. The screenshot is grouped by service; a project filter is not visible. This report uses the confirmed activity scope, without claiming a resource-by-resource allocation of shared storage or background charges.

This is the cost displayed at capture time, not a guarantee of a final zero bill. Forecast values are excluded. Later billing updates and retained SQL storage or secrets can change subsequent charges.

Lifecycle commands for reproduction

```
python3 infra.py finish          # temporary SQL/VM stop; retain deployment
bash cleanup.sh --plan           # inspect teardown scope
bash cleanup.sh                  # stop SQL; remove owned temporary resources
```

Before teardown, tools/collect_cloud_evidence.py --run-id RUN_ID --stop collects audit/metric evidence for the completed run and stops SQL and the VMs. Substitute the actual completed run ID. SQL data must not be deleted to simplify screenshots.

Evidence and implementation references

Saved curl commands, responses and complete table snapshots are in report/evidence/demo-8131a0d413fb4069/. Formal commands, exits and statistics are in report/evidence/load-20261021T075735Z-5b3f60c2/. The full gzip SQL export is retained locally under results/load-20261021T075735Z-5b3f60c2/ and excluded from routine Git staging; its hash appears on page 4.

Independent checks: formal-run-review.json, post-load-review.json and cleanup-verification.json in report/evidence/. Original screenshot hashes: report/screenshots/manifest.json. Billing image and owner scope: report/evidence/billing/ and billing-review.json.

Google Cloud: Cloud SQL Python Connector; start/stop Cloud SQL; authenticated Scheduler; Pub/Sub monitoring; billing reports. Assignment: instructor-supplied CS528-HW_5.pdf.

Appendix A1. HTTP 200: existing file

Figure 1. complete tables before the request.

Local replay of saved evidence from the real October 21 curl demonstration. Screenshot captured October 22; no new request was executed.

```
PAGE 1/9 | Suggested screenshot: success-before.png
SAVED REAL EVIDENCE REPLAY - SUCCESS
Run: demo-8131a0d413fb4069 | Original capture: 2026-10-21T06:44:26.386498+00:00
Read-only local display. No curl execution, cloud query or database change.

FULL TABLES BEFORE
requests: ALL 1 rows, ALL 15 columns
  Record 1
  request_id="53b3df6f-3f4c-49f4-8a81-91d12b8fcce5" | request_time="2026-10-21 0
4:58:30.389453"
  run_id=null | method="GET" | requested_file=""
  country=null | client_ip="45.198.224.125" | peer_ip="45.198.224.125"
  gender=null | age=null | income=null | is_banned=false | http_status=404
  client_time=null | time_of_day=null

failed_requests: ALL 1 rows, ALL 3 columns
  Record 1: request_time="2026-10-21 04:58:30.389453" | requested_file="" | erro
r_code=404
```

Initial state: 1 request row and 1 error row. The preexisting public 404 request is retained.

Appendix A2. HTTP 200: existing file

Figure 2. saved curl command and response.

Local replay of saved evidence from the real October 21 curl demonstration. Screenshot captured October 22; no new request was executed.

```
PAGE 2/9 | Suggested screenshot: success-response.png
SAVED REAL EVIDENCE REPLAY - SUCCESS
Run: demo-8131a0d413fb4069 | Original capture: 2026-10-21T06:44:26.386498+00:00
Read-only local display. No curl execution, cloud query or database change.

ORIGINAL EXECUTED COMMAND (display only; historical IP was released)
curl --silent --show-error --max-time 60 --dump-header
  /Users/thomasfang/CS528/CS528-HWs/CS528-HW5/report/evidence/demo-8131a0d413fb4
069/success-headers.txt --output
  /Users/thomasfang/CS528/CS528-HWs/CS528-HW5/report/evidence/demo-8131a0d413fb4
069/success-body.txt --write-out
  '%{http_code}' -H 'X-country: United States' -H 'X-client-IP: 192.0.2.25' -H '
X-gender: Female' -H 'X-age:
  26-35' -H 'X-income: 40k-60k' -H 'X-time: 2026-10-20 13:00:00'
  http://35.193.157.51:8080/__runs/demo-8131a0d413fb4069/data/0.html

SAVED RESPONSE HEADERS
HTTP/1.1 200 OK
Server: BaseHTTP/0.6 Python/3.11.2
Date: Wed, 21 Oct 2026 06:44:24 GMT
Content-Type: text/html
Content-Length: 97586
Cache-Control: no-store
Connection: close


Saved response body: 97586 bytes
Body SHA-256: c3be7cacaad7e54bac9e5050419002b0545418e31094933b4239813e2a6f2b88
Verified changes: +1 request, +0 error row(s).
```

Real response: HTTP 200 OK; 97,586 bytes; +1 request and +0 error rows.

Appendix A3. HTTP 200: existing file

Figure 3. complete tables after the request.

Local replay of saved evidence from the real October 21 curl demonstration. Screenshot captured October 22; no new request was executed.

```
PAGE 3/9 | Suggested screenshot: success-after.png
SAVED REAL EVIDENCE REPLAY - SUCCESS
Run: demo-8131a0d413fb4069 | Original capture: 2026-10-21T06:44:26.386498+00:00
Read-only local display. No curl execution, cloud query or database change.

FULL TABLES AFTER
requests: ALL 2 rows, ALL 15 columns
  Record 1
  request_id="53b3df6f-3f4c-49f4-8a81-91d12b8fcce5" | request_time="2026-10-21 0
4:58:30.389453"
  run_id=null | method="GET" | requested_file=""
  country=null | client_ip="45.198.224.125" | peer_ip="45.198.224.125"
  gender=null | age=null | income=null | is_banned=false | http_status=404
  client_time=null | time_of_day=null
  Record 2
  request_id="b492792f-3474-42f3-9662-0da1f5e2e784" | request_time="2026-10-21 0
6:44:23.565363"
  run_id="demo-8131a0d413fb4069" | method="GET" | requested_file="data/0.html"
  country="United States" | client_ip="192.0.2.25" | peer_ip="52.124.36.217"
  gender="Female" | age="26-35" | income="40k-60k" | is_banned=false | http_stat
us=200
  client_time="2026-10-20 13:00:00" | time_of_day="13:00:00"

failed_requests: ALL 1 rows, ALL 3 columns
  Record 1: request_time="2026-10-21 04:58:30.389453" | requested_file="" | erro
r_code=404
```

After success: 2 request rows and 1 error row. The new United States request has HTTP 200.

Appendix A4. HTTP 404: missing file

Figure 4. complete tables before the request.

Local replay of saved evidence from the real October 21 curl demonstration. Screenshot captured October 22; no new request was executed.

```
PAGE 4/9 | Suggested screenshot: missing-before.png
SAVED REAL EVIDENCE REPLAY - MISSING
Run: demo-8131a0d413fb4069 | Original capture: 2026-10-21T06:44:26.386498+00:00
Read-only local display. No curl execution, cloud query or database change.

FULL TABLES BEFORE
requests: ALL 2 rows, ALL 15 columns
  Record 1
  request_id="53b3df6f-3f4c-49f4-8a81-91d12b8fcce5" | request_time="2026-10-21 0
4:58:30.389453"
  run_id=null | method="GET" | requested_file=""
  country=null | client_ip="45.198.224.125" | peer_ip="45.198.224.125"
  gender=null | age=null | income=null | is_banned=false | http_status=404
  client_time=null | time_of_day=null
  Record 2
  request_id="b492792f-3474-42f3-9662-0da1f5e2e784" | request_time="2026-10-21 0
6:44:23.565363"
  run_id="demo-8131a0d413fb4069" | method="GET" | requested_file="data/0.html"
  country="United States" | client_ip="192.0.2.25" | peer_ip="52.124.36.217"
  gender="Female" | age="26-35" | income="40k-60k" | is_banned=false | http_stat
us=200
  client_time="2026-10-20 13:00:00" | time_of_day="13:00:00"

failed_requests: ALL 1 rows, ALL 3 columns
  Record 1: request_time="2026-10-21 04:58:30.389453" | requested_file="" | erro
r_code=404
```

Before the missing-file request: 2 request rows and 1 error row; identical to the prior after-snapshot.

Appendix A5. HTTP 404: missing file

Figure 5. saved curl command and response.

Local replay of saved evidence from the real October 21 curl demonstration. Screenshot captured October 22; no new request was executed.

```
PAGE 5/9 | Suggested screenshot: missing-response.png
SAVED REAL EVIDENCE REPLAY - MISSING
Run: demo-8131a0d413fb4069 | Original capture: 2026-10-21T06:44:26.386498+00:00
Read-only local display. No curl execution, cloud query or database change.

ORIGINAL EXECUTED COMMAND (display only; historical IP was released)
curl --silent --show-error --max-time 60 --dump-header
  /Users/thomasfang/CS528/CS528-HWs/CS528-HW5/report/evidence/demo-8131a0d413fb4
069/missing-headers.txt --output
  /Users/thomasfang/CS528/CS528-HWs/CS528-HW5/report/evidence/demo-8131a0d413fb4
069/missing-body.txt --write-out
  '%{http_code}' -H 'X-country: United States' -H 'X-client-IP: 192.0.2.25' -H '
X-gender: Female' -H 'X-age:
  26-35' -H 'X-income: 40k-60k' -H 'X-time: 2026-10-20 13:00:00'
  http://35.193.157.51:8080/__runs/demo-8131a0d413fb4069/data/hw5-file-that-does
-not-exist.html

SAVED RESPONSE HEADERS
HTTP/1.1 404 Not Found
Server: BaseHTTP/0.6 Python/3.11.2
Date: Wed, 21 Oct 2026 06:44:25 GMT
Content-Type: text/plain; charset=utf-8
Content-Length: 16
Cache-Control: no-store
Connection: close


Saved response body: 16 bytes
Body SHA-256: 8a482f2271a42c5f54c96e816a84340a6f2357a5b81f927d07d00788f5140a41
Verified changes: +1 request, +1 error row(s).
Saved error body: "File not found.\n"
```

Real response: HTTP 404 Not Found; +1 request and +1 error row.

Appendix A6. HTTP 404: missing file

Figure 6. complete tables after the request.

Local replay of saved evidence from the real October 21 curl demonstration. Screenshot captured October 22; no new request was executed.

```
PAGE 6/9 | Suggested screenshot: missing-after.png
SAVED REAL EVIDENCE REPLAY - MISSING
Run: demo-8131a0d413fb4069 | Original capture: 2026-10-21T06:44:26.386498+00:00
Read-only local display. No curl execution, cloud query or database change.

FULL TABLES AFTER
requests: ALL 3 rows, ALL 15 columns
  Record 1
  request_id="53b3df6f-3f4c-49f4-8a81-91d12b8fcce5" | request_time="2026-10-21 0
4:58:30.389453"
  run_id=null | method="GET" | requested_file=""
  country=null | client_ip="45.198.224.125" | peer_ip="45.198.224.125"
  gender=null | age=null | income=null | is_banned=false | http_status=404
  client_time=null | time_of_day=null
  Record 2
  request_id="b492792f-3474-42f3-9662-0da1f5e2e784" | request_time="2026-10-21 0
6:44:23.565363"
  run_id="demo-8131a0d413fb4069" | method="GET" | requested_file="data/0.html"
  country="United States" | client_ip="192.0.2.25" | peer_ip="52.124.36.217"
  gender="Female" | age="26-35" | income="40k-60k" | is_banned=false | http_stat
us=200
  client_time="2026-10-20 13:00:00" | time_of_day="13:00:00"
  Record 3
  request_id="73abace9-b36e-4934-bc39-71c94b8bf5df" | request_time="2026-10-21 0
6:44:24.943455"
  run_id="demo-8131a0d413fb4069" | method="GET" | requested_file="data/hw5-file-
that-does-not-exist.html"
  country="United States" | client_ip="192.0.2.25" | peer_ip="52.124.36.217"
  gender="Female" | age="26-35" | income="40k-60k" | is_banned=false | http_stat
us=404
  client_time="2026-10-20 13:00:00" | time_of_day="13:00:00"

failed_requests: ALL 2 rows, ALL 3 columns
  Record 1: request_time="2026-10-21 04:58:30.389453" | requested_file="" | erro
r_code=404
  Record 2: request_time="2026-10-21 06:44:24.943455" | requested_file="data/hw5
-file-that-does-not-exist.html" | error_code=404
```

After the missing-file request: 3 request rows and 2 error rows. The new 404 record matches by time and path.

Appendix A7. HTTP 400: forbidden country

Figure 7. complete tables before the request.

Local replay of saved evidence from the real October 21 curl demonstration. Screenshot captured October 22; no new request was executed.

```
PAGE 7/9 | Suggested screenshot: banned-before.png
SAVED REAL EVIDENCE REPLAY - BANNED
Run: demo-8131a0d413fb4069 | Original capture: 2026-10-21T06:44:26.386498+00:00
Read-only local display. No curl execution, cloud query or database change.

FULL TABLES BEFORE
requests: ALL 3 rows, ALL 15 columns
  Record 1
  request_id="53b3df6f-3f4c-49f4-8a81-91d12b8fcce5" | request_time="2026-10-21 0
4:58:30.389453"
  run_id=null | method="GET" | requested_file=""
  country=null | client_ip="45.198.224.125" | peer_ip="45.198.224.125"
  gender=null | age=null | income=null | is_banned=false | http_status=404
  client_time=null | time_of_day=null
  Record 2
  request_id="b492792f-3474-42f3-9662-0da1f5e2e784" | request_time="2026-10-21 0
6:44:23.565363"
  run_id="demo-8131a0d413fb4069" | method="GET" | requested_file="data/0.html"
  country="United States" | client_ip="192.0.2.25" | peer_ip="52.124.36.217"
  gender="Female" | age="26-35" | income="40k-60k" | is_banned=false | http_stat
us=200
  client_time="2026-10-20 13:00:00" | time_of_day="13:00:00"
  Record 3
  request_id="73abace9-b36e-4934-bc39-71c94b8bf5df" | request_time="2026-10-21 0
6:44:24.943455"
  run_id="demo-8131a0d413fb4069" | method="GET" | requested_file="data/hw5-file-
that-does-not-exist.html"
  country="United States" | client_ip="192.0.2.25" | peer_ip="52.124.36.217"
  gender="Female" | age="26-35" | income="40k-60k" | is_banned=false | http_stat
us=404
  client_time="2026-10-20 13:00:00" | time_of_day="13:00:00"

failed_requests: ALL 2 rows, ALL 3 columns
  Record 1: request_time="2026-10-21 04:58:30.389453" | requested_file="" | erro
r_code=404
  Record 2: request_time="2026-10-21 06:44:24.943455" | requested_file="data/hw5
-file-that-does-not-exist.html" | error_code=404
```

Before the Iran request: 3 request rows and 2 error rows; identical to the prior after-snapshot.

Appendix A8. HTTP 400: forbidden country

Figure 8. saved curl command and response.

Local replay of saved evidence from the real October 21 curl demonstration. Screenshot captured October 22; no new request was executed.

```
PAGE 8/9 | Suggested screenshot: banned-response.png
SAVED REAL EVIDENCE REPLAY - BANNED
Run: demo-8131a0d413fb4069 | Original capture: 2026-10-21T06:44:26.386498+00:00
Read-only local display. No curl execution, cloud query or database change.

ORIGINAL EXECUTED COMMAND (display only; historical IP was released)
curl --silent --show-error --max-time 60 --dump-header
  /Users/thomasfang/CS528/CS528-HWs/CS528-HW5/report/evidence/demo-8131a0d413fb4
069/banned-headers.txt --output
  /Users/thomasfang/CS528/CS528-HWs/CS528-HW5/report/evidence/demo-8131a0d413fb4
069/banned-body.txt --write-out
  '%{http_code}' -H 'X-country: Iran' -H 'X-client-IP: 192.0.2.25' -H 'X-gender:
 Female' -H 'X-age: 26-35' -H
  'X-income: 40k-60k' -H 'X-time: 2026-10-20 13:00:00'
  http://35.193.157.51:8080/__runs/demo-8131a0d413fb4069/data/0.html

SAVED RESPONSE HEADERS
HTTP/1.1 400 Bad Request
Server: BaseHTTP/0.6 Python/3.11.2
Date: Wed, 21 Oct 2026 06:44:25 GMT
Content-Type: text/plain; charset=utf-8
Content-Length: 38
Cache-Control: no-store
Connection: close


Saved response body: 38 bytes
Body SHA-256: f155e6450c546f22ad5168e6ad3e15891dd05466767b52bff41b8579d9cd541b
Verified changes: +1 request, +1 error row(s).
Saved error body: "Permission denied: request from Iran.\n"
```

Real response: HTTP 400 Bad Request; country Iran; +1 request and +1 error row.

Appendix A9. HTTP 400: forbidden country

Figure 9. complete tables after the request.

Local replay of saved evidence from the real October 21 curl demonstration. Screenshot captured October 22; no new request was executed.

```
PAGE 9/9 | Suggested screenshot: banned-after.png
SAVED REAL EVIDENCE REPLAY - BANNED
Run: demo-8131a0d413fb4069 | Original capture: 2026-10-21T06:44:26.386498+00:00
Read-only local display. No curl execution, cloud query or database change.

FULL TABLES AFTER
requests: ALL 4 rows, ALL 15 columns
  Record 1
  request_id="53b3df6f-3f4c-49f4-8a81-91d12b8fcce5" | request_time="2026-10-21 04:58:30.389453"
  run_id=null | method="GET" | requested_file=""
  country=null | client_ip="45.198.224.125" | peer_ip="45.198.224.125"
  gender=null | age=null | income=null | is_banned=false | http_status=404
  client_time=null | time_of_day=null
  Record 2
  request_id="b492792f-3474-42f3-9662-0da1f5e2e784" | request_time="2026-10-21 06:44:23.565363"
  run_id="demo-8131a0d413fb4069" | method="GET" | requested_file="data/0.html"
  country="United States" | client_ip="192.0.2.25" | peer_ip="52.124.36.217"
  gender="Female" | age="26-35" | income="40k-60k" | is_banned=false | http_status=200
  client_time="2026-10-20 13:00:00" | time_of_day="13:00:00"
  Record 3
  request_id="73abace9-b36e-4934-bc39-71c94b8bf5df" | request_time="2026-10-21 06:44:24.943455"
  run_id="demo-8131a0d413fb4069" | method="GET" | requested_file="data/hw5-file-that-does-not-exist.html"
  country="United States" | client_ip="192.0.2.25" | peer_ip="52.124.36.217"
  gender="Female" | age="26-35" | income="40k-60k" | is_banned=false | http_status=404
  client_time="2026-10-20 13:00:00" | time_of_day="13:00:00"
  Record 4
  request_id="1f0a52b6-c055-4648-84d9-f63fd9974c61" | request_time="2026-10-21 06:44:25.758649"
  run_id="demo-8131a0d413fb4069" | method="GET" | requested_file="data/0.html"
  country="Iran" | client_ip="192.0.2.25" | peer_ip="52.124.36.217"
  gender="Female" | age="26-35" | income="40k-60k" | is_banned=true | http_status=400
  client_time="2026-10-20 13:00:00" | time_of_day="13:00:00"

failed_requests: ALL 3 rows, ALL 3 columns
  Record 1: request_time="2026-10-21 04:58:30.389453" | requested_file="" | error_code=404
  Record 2: request_time="2026-10-21 06:44:24.943455" | requested_file="data/hw5-file-that-does-not-exist.html" | error_code=404
  Record 3: request_time="2026-10-21 06:44:25.758649" | requested_file="data/0.html" | error_code=400
```

After the Iran request: 4 request rows and 3 error rows. is_banned=true and error_code=400 are recorded.

Appendix B. Billing snapshot: October 20-22, 2026

Displayed total: **$0.00** | Savings: **$1.61** | Captured October 22, 05:41 local time. Forecast excluded.

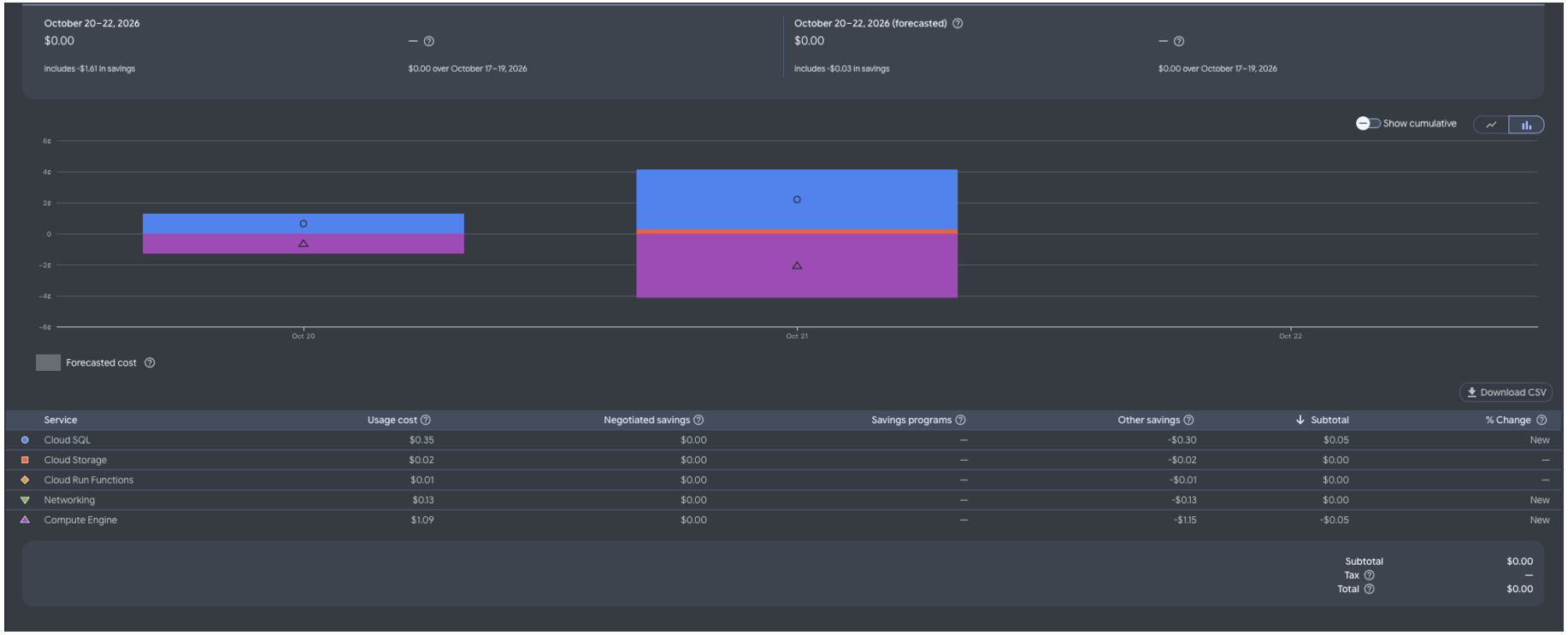


Figure 10. Original service-grouped billing screenshot. An enlarged transcription of all service rows appears on page 8. The original image is embedded without resampling so details can be inspected by zooming.

The owner confirmed no gcloud operations on other projects during the displayed interval. The project filter is not visible in the screenshot. Values describe the report at capture time and can change with delayed billing or continued retained-storage charges.